Multi-Component Style Vectors with Norm-Preserving Steering for Personalized Text Generation

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I. Project Details

We are following the criterion: **"Reproduce paper results if code not available"**, combined with proposing and evaluating an improvement over the base method. The project builds on *StyleVector: Personalized Text Generation with Contrastive Activation Steering* [1] (ACL 2025), which represents a user's writing style as a single linear direction (a "style vector") in an LLM's activation space and steers generation towards that style at inference time, without any retrieval or fine-tuning. As of the writing of this proposal, no official public code repository has been released by the authors. We will therefore (1) reimplement the StyleVector pipeline from scratch following the algorithm and equations described in the paper, (2) reproduce a subset of their reported results, and (3) implement and evaluate a proposed extension that directly targets a limitation the authors themselves acknowledge.

Team Composition and Individual Contributions:

- **Member 1:** Slok Tulsyan, 12342080, `slok@iitbhilai.ac.in`. Since this is a single-member team, I am solely responsible for all parts of the project: literature review, reimplementation of the baseline StyleVector pipeline, design and implementation of the proposed multi-component/norm-preserving extension, running all experiments, evaluation, ablations, and the final report/presentation.

Note: As a solo team, all components listed under Methodology and Expected Outcomes below are my individual responsibility, sequenced across the semester as detailed in the timeline in Section V.

II. Problem Statement

Context and motivation. Large language models are typically optimized to produce a single, average, "one-size-fits-all" writing style. In practice, users want AI-generated text (emails, reviews, headlines, social posts) that sounds like *them*. Existing personalization approaches fall into two families: Retrieval-Augmented Generation (RAG), which stuffs a few of the user's past texts into the prompt, and Parameter-Efficient Fine-Tuning (PEFT), which trains a small per-user adapter (e.g., LoRA). Both approaches entangle *what* the user is writing about (content) with *how* they write (style), and both scale poorly: RAG's retrieval latency grows with history size, and PEFT requires storing and loading separate parameters for every user.

Why this matters. As personalized AI assistants become mainstream (writing assistants, customer-support agents, recommendation explanations), the ability to cheaply and accurately capture an individual's style – without per-user training or large storage – is a practically important and actively studied problem in the ML/NLP community.

Scope of our work. StyleVector addresses scalability by encoding a user's style as a single contrastively-derived vector in activation space and adding it to hidden states during generation (Eq. 1 below). However, the authors' own *Limitations* section notes that a single vector likely *conflates* multiple, independent stylistic dimensions (lexical choice, syntactic structure, punctuation habits, discourse-level patterns) into one linear direction, and that steering strength is sensitive: too weak has no effect, too strong destabilizes generation. Our project's scope is narrow and well-defined: (a) reproduce the core StyleVector method (Mean-Difference extraction + single-layer steering) on 2–3 tasks from the LaMP benchmark [2], and (b) design, implement, and evaluate an extension – *multi-component style vectors* (via top-k PCA over contrastive activation differences) combined with *norm-preserving steering* – that directly targets the two limitations above.

III. Methodology

Background (baseline being reproduced). For a user u with history $P_u = \{(x_i, y_i)\}_{i=1}^{|P_u|}$, StyleVector:

1) Generates a style-agnostic response $\hat{y}_i = M_g(x_i)$ for each historical input using a general-purpose LLM M_g.
2) Extracts contrastive activations at a chosen layer ℓ: $a^\ell_{p,i} = h_\ell(x_i \oplus y_i)$ (authentic) and $a^\ell_{n,i} = h_\ell(x_i \oplus \hat{y}_i)$ (generic), where $h_\ell(\cdot)$ is the last-token hidden state at layer ℓ.
3) Computes a single style vector via the mean difference: $s^\ell_u = \frac{1}{|P_u|}\sum_i (a^\ell_{p,i} - a^\ell_{n,i})$.
4) Steers generation by adding the scaled vector to every generated token's hidden state: $h'_\ell(x)_t = h_\ell(x)_t + \alpha\, s^\ell_u$.

Our proposed extension. We generalize Steps 3–4 as follows.

(a) Multi-component extraction. Instead of a single direction, we compute the top-k principal components $\{v_1, \ldots, v_k\}$ of the contrastive difference set $\{\Delta_i\} \cup \{-\Delta_i\}$, $\Delta_i = a^\ell_{p,i} - a^\ell_{n,i}$, via eigendecomposition of $\sum_i (\Delta_i \Delta_i^\top + (-\Delta_i)(-\Delta_i)^\top)$ (an extension of the paper's own PCA variant, Eq. 10, from $k{=}1$ to general k). Each component captures a different, hopefully more disentangled, axis of stylistic variation (e.g.,

lexical vs. structural), addressing the "single-vector conflation" limitation directly.

(b) Norm-preserving multi-component steering. We steer with a weighted combination of components, scaled by $\alpha_j \propto \sqrt{\lambda_j}$ (their explained variance), and then renormalize the resulting activation back to its original magnitude:

$$h'_\ell(x)_t = \|h_\ell(x)_t\| \cdot \frac{h_\ell(x)_t + \sum_{j=1}^{k} \alpha_j v_j}{\left\| h_\ell(x)_t + \sum_{j=1}^{k} \alpha_j v_j \right\|}. \quad (1)$$

This is intended to prevent the activation-magnitude blow-up we observed in the paper's own strength-sweep results (their Fig. 3), where large α degrades generation quality.

Frameworks/tools. We will use HuggingFace Transformers for model loading and forward-hook-based activation extraction/injection, PyTorch for the linear algebra (eigendecomposition/SVD via `torch.linalg`), and standard NLP evaluation libraries for ROUGE-L and METEOR, matching the original paper's protocol so results are directly comparable.

A schematic of the overall pipeline, showing where our extension modifies the original StyleVector framework, is shown in Fig. 1.

IV. Resource and Dataset Availability

Datasets. We will use the publicly available **LaMP** benchmark [2] (https://lamp-benchmark.github.io/), specifically the *News Headline Generation*, *Scholarly Title Generation*, and *Tweet Paraphrasing* tasks – these are the same tasks used in the original paper and have the smallest per-user history sizes, making them the fastest to iterate on. Each task provides, per user, a set of historical (input, output) text pairs and held-out test queries with ground-truth outputs. Given compute constraints as a solo project, we will subsample ∼150–250 test users per task (versus the full test split) for development and ablations, and report on the full split only for final numbers if time permits.

Preprocessing. Minimal preprocessing is required: tokenization is handled by the chosen LLM's tokenizer; we will truncate very long histories to a fixed cap to bound the number of forward passes per user.

Why this dataset is suitable. LaMP is the exact benchmark the original paper uses, is small enough for solo-project timelines, is pre-split into per-user train/val/test, and already includes standard evaluation scripts for ROUGE-L/METEOR.

Compute/hardware. A single consumer/lab GPU with ≥16GB VRAM (e.g., RTX 3090/4090 or a Colab/Kaggle A100 instance) should suffice if we use a smaller open model (see below) with 8-bit quantization.

Software. Python, PyTorch, HuggingFace `transformers` and `accelerate`, `bitsandbytes` (8-bit loading), `scikit-learn` (PCA sanity checks), `rouge-score` and `nltk` (METEOR), `matplotlib`/`seaborn` for plots.

Model. Due to compute limits (vs. the original paper's LLaMA-2-7B on 8×RTX 3090), we plan to use a smaller open instruction-tuned model such as **Qwen2.5-1.5B/3B-Instruct** or **Llama-3.2-3B-Instruct** for both the steered model M and, optionally, a larger API/local model as the generic response generator M_g. If additional GPU access becomes available (department cluster / Kaggle P100/T4), we will scale up to a 7B model to more closely match the original setup.

Fallback if resources are limited. If GPU access is constrained, we will (i) restrict experiments to the single smallest task (Tweet Paraphrasing), (ii) reduce the number of PCA components tested, and (iii) rely on 8-bit quantized 1.5B–3B models throughout.

V. Limitations Found

The original StyleVector paper itself identifies several limitations that motivate our project:

- **Single-vector conflation:** one direction per user may mix lexical, syntactic, and discourse-level style signals that would ideally be disentangled and controlled separately.
- **Steering instability:** the reported strength sweep (their Fig. 3) shows that generation quality collapses for large α, suggesting the naive additive intervention pushes activations outside their normal operating range.
- **Domain homogeneity assumption:** the benchmarks assume one consistent style per user across all their history, which the authors flag as unrealistic (users may write differently in different contexts); we will not attempt to fully solve this in our scope but will note it as future work.

How we address these. Our top-k PCA extraction directly targets the first limitation by keeping multiple, separately-weighted style directions instead of collapsing everything into one mean-difference vector. Our norm-preserving projection directly targets the second limitation by decoupling the *direction* of the steering perturbation from its effect on activation *magnitude*, which we hypothesize will allow larger, more effective steering strengths without the quality collapse observed in the original paper.

What differentiates our work. Unlike the original paper, which only evaluates a single steering direction and a fixed additive intervention, we introduce (a) a tunable multi-component representation of style and (b) a magnitude-safe steering rule, and we run a controlled 2×2 ablation (single- vs. multi-component $\times$ with/without norm preservation) to isolate which piece of our contribution actually drives any improvement – something the original paper does not do.

VI. Expected Outcomes

Performance metrics. We aim to (i) reproduce the original paper's reported ROUGE-L/METEOR trends for StyleVector vs. a non-personalized baseline within a reasonable margin (allowing for our use of a smaller model), and (ii) show a measurable improvement in ROUGE-L/METEOR from our multi-component + norm-preserving extension over our reproduced single-vector baseline, on at least 2 of the 3 chosen LaMP tasks.

Ablations. A full 2×2 ablation table (single vs. multi-component; with vs. without norm preservation), plus a sweep over the number of components $k \in \{1, 3, 5\}$ and steering layer ℓ.

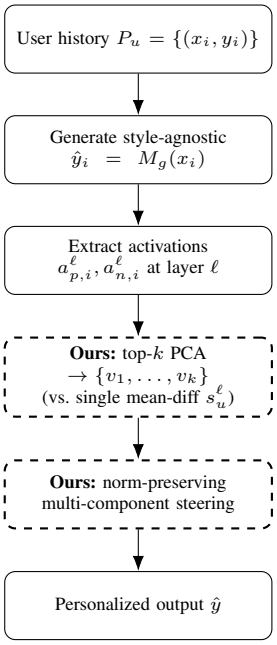


Fig. 1. Proposed pipeline. Solid boxes: original StyleVector stages. Dashed boxes: our proposed extension (multi-component PCA extraction and norm-preserving steering).

TABLE I
PLANNED ABLATION GRID TO ISOLATE CONTRIBUTIONS

	No norm preservation	With norm preservation
Single vector (k=1)	Original StyleVector	Ablation
Multi-component (k>1)	Ablation	**Our full method**

Qualitative analysis. A case study (in the style of the original paper's Figure 5) showing example generations under the baseline vs. our method for a specific test user, plus a t-SNE visualization of the extracted multi-component style directions.

Final deliverables. (1) A reproducible codebase implementing both the baseline StyleVector pipeline and our extension; (2) a final project report documenting methodology, results, and ablations; (3) a short presentation/demo comparing baseline vs. improved steering on a handful of live examples.

REFERENCES

[1] J. Zhang, Y. Liu, W. Wang, Q. Liu, S. Wu, L. Wang, and T.-S. Chua, "Personalized text generation with contrastive activation steering," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2025, pp. 7128–7141.

[2] A. Salemi, S. Mysore, M. Bendersky, and H. Zamani, "Lamp: When large language models meet personalization," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2024, pp. 7370–7392.

[3] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.

[4] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

APPENDIX

An AI assistant (Anthropic's Claude) was used during the ideation phase of this proposal: to help summarize and explain the StyleVector paper's methodology and equations for background understanding, to brainstorm candidate directions for improving upon the paper's stated limitations, and to help structure this proposal document into the required template format. All final decisions on the specific proposed method (multi-component PCA style vectors with norm-preserving steering), scope, dataset/task selection, and experimental plan were made by the author. No external code repository was used; the reimplementation will be written from scratch based on the equations and algorithm described in the original paper [1].